COMPATIBILITY OF THE CYCLE INTERSECTION WITH FLAT BASE CHANGE AND FLAT PULLBACK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Chow Homology Situation setting up base change of cycles with a morphism g : S' to S and a function c on the delta dimensions (chow chapter, Situation chow-situation-setup-base-change, canonical tag 0FVG). cited at proof line 2265. Fixes the base-change setup: delta' = delta + c and the meaning of the base change of a family of cycles.
- Chow Homology section on change of base for cycles, in particular that the pullback of a prime cycle beta is the sum over irreducible components with multiplicities (chow chapter, Section chow-section-change-base, canonical tag 0FVF). cited at proof line 2272. Source of the formula beta' = sum n_a [Z'_a] used in the multiplicity computation.
- Lemma allowing the intersection operation to be checked locally on Y: for open U in X and V in Y with f(U) in V, restricting the factors restricts the intersection (more precisely the relative-cycles chapter, Lemma lemma-action-opens, canonical tag 0H67; rejected as a standalone card, cited here as a named input). cited at proof line 2279. Licences the reduction to Y affine so that beta is a finite linear combination of prime cycles.
- Bilinearity of the intersection operation in the second variable (relative-cycles chapter, Lemma lemma-action-bilinear, canonical tag 0H66; rejected as a standalone card, cited here as a named input). cited at proof line 2282. Reduces the equality from arbitrary beta to the prime-cycle case beta = [Z].
- Commutativity of the square relating base change of zero-cycles on the generic fibre and flat pullback (chow chapter, Lemma chow-lemma-pullback-base-change-pullback, canonical tag 0FVK). cited at proof line 2344. The named endpoint: the diagram on cycle groups base-changed by the generic point and pulled back flat is commutative, which is exactly the component and multiplicity claim after the birational transport.

Source reference. Original reference 0FVG: sources/stacks-chow/tags.txt, source lines 19811–19811. Cited at proof line 2265: fixes the base-change setup (delta prime equals delta plus c) for the whole proof.

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Source reference. Original section 0FVF: sources/stacks-chow/tags.txt, source lines 19810–19810. Cited at proof line 2272: source of the pullback formula for beta with component multiplicities.

Source reference. Original lemma 0H67: sources/stacks-chow/tags.txt, source lines 21412–21412. Cited at proof line 2279: licences working locally on Y so beta splits into prime cycles.

Source reference. Original lemma 0H66: sources/stacks-chow/tags.txt, source lines 21411–21411. Cited at proof line 2282: reduces the equality to the prime-cycle case.

Source reference. Original lemma 0FVK: sources/stacks-chow/tags.txt, source lines 19815–19815. Cited at proof line 2344: the named commutative square that finishes the multiplicity computation.

Lemma 0.2. *Forming $\alpha \cap \beta$ is compatible with flat base change and flat pullback (see proof for elucidation).*

Proof. Let (S, δ), (S', δ'), $g : S' \to S$, and $c \in \mathbf{Z}$ be as in Chow Homology, Situation 0FVG. Let $X \to Y$ be a morphism of schemes locally of finite type over S. Denote $X' \to Y'$ the base change of $X \to Y$ by g. Let α be a family of r-cycles on the fibres of X/Y. Let $\beta \in Z_e(Y)$. Denote α' the base change of α by $Y' \to Y$. Denote $\beta' = g^*\beta \in Z_{e+c}(Y')$ the pullback of β by g, see Chow Homology, Section 0FVF. Compatibility with base change means $\alpha' \cap \beta'$ is the base change of $\alpha \cap \beta$.

Proof of compatibility with base change. Since we are proving an equality of cycles on X', we may work locally on Y, see Lemma 0H67. Thus we may assume Y is affine. In particular β is a finite linear combination of prime cycles. Since $- \cap -$ is linear in the second variable (Lemma 0H66), it suffices to prove the equality when $\beta = [Z]$ for some integral closed subscheme $Z \subset Y$ of δ-dimension e.

Let $y \in Z$ be the generic point. Write $\alpha_y = \sum m_j [V_j]$. Let $\overline{V}_j$ be the closure of V_j in X. Then we have

$$\alpha \cap \beta = \sum m_j [\overline{V}_j]$$

The base change of β is $\beta' = \sum [Z \times_S S']_{e+c}$ as a cycle on $Y' = Y \times_S S'$. Let $Z'_a \subset Z \times_S S'$ be the irreducible components, denote $y'_a \in Z'_a$ their generic points, and denote n_a the multiplicity of Z'_a in $Z \times_S S'$. We have

$$\beta' = \sum [Z \times_S S']_{e+c} = \sum n_a [Z'_a]$$

We have $\alpha'_{y'_a} = \sum m_j [V_{j,\kappa(y'_a)}]_r$ because α' is the base change of α by $Y' \to Y$. Let $V'_{jab} \subset V_{j,\kappa(y'_a)}$ be the irreducible components and denote m_{jab} the multiplicity of V'_{jab} in $V_{j,\kappa(y'_a)}$. We have

$$\alpha'_{y'_a} = \sum m_j [V_{j,\kappa(y'_a)}]_r = \sum m_j m_{jab} [V'_{jab}]$$

Thus we we have

$$\alpha' \cap \beta' = \sum n_a m_j m_{jab} [\overline{V}'_{jab}]$$

where $\overline{V}'_{jab}$ is the closure of V'_{jab} in X'. Thus to prove the desired equality it suffices to prove

(1) the irreducible components of $\overline{V}_j \times_S S'$ are the schemes $\overline{V}'_{jab}$ and
(2) the multiplicity of $\overline{V}'_{jab}$ in $\overline{V}_j \times_S S'$ is equal to $n_a m_{jab}$.

Note that $V_j \to \overline{V}_j$ is a birational morphism of integral schemes. The morphisms $V_j \times_S S' \to V_j$ and $\overline{V}_j \times_S S' \to \overline{V}_j$ are flat and hence map generic points of irreducible components to the (unique) generic points of V_j and $\overline{V}_j$. It follows that $V_j \times_S S' \to \overline{V}_j \times_S S'$ is a birational morphisms hence induces a bijection on irreducible components and identifies their multiplicities. This means that it suffices to prove that the irreducible components of $V_j \times_S S'$ are the schemes V'_{jab} and the multiplicity of V'_{jab} in $V_j \times_S S'$ is equal to $n_a m_{jab}$. However, then we are just saying that the diagram

$$\begin{array}{ccc} Z_r(V_j) & \longrightarrow & Z_{r+c}(V_j \times_S S') \\ \uparrow & & \uparrow \\ Z_0(\mathrm{Spec}(\kappa(y))) & \longrightarrow & Z_c(\mathrm{Spec}(\kappa(y)) \times_S S') \end{array}$$

is commutative where the horizontal arrows are base change by $\mathrm{Spec}(\kappa(y)) \times_S S' \to \mathrm{Spec}(\kappa(y))$ and the vertical arrows are flat pullback. This was shown in Chow Homology, Lemma 0FVK.

The statement in the lemma on flat pullback means the following. Let (S, δ), $X \to Y$, α, and β be as in the construction of $\alpha \cap \beta$ above. Let $Y' \to Y$ be a flat morphism, locally of finite type, and of relative dimension c. Then we can let α' be the base change of α by $Y' \to Y$ and β' the flat pullback of β. Compatibility with flat pullback means $\alpha' \cap \beta'$ is the flat pullback of $\alpha \cap \beta$ by $X \times_Y Y' \to Y$. This is actually a special case of the discussion above if we set $S = Y$ and $S' = Y'$. $\square$